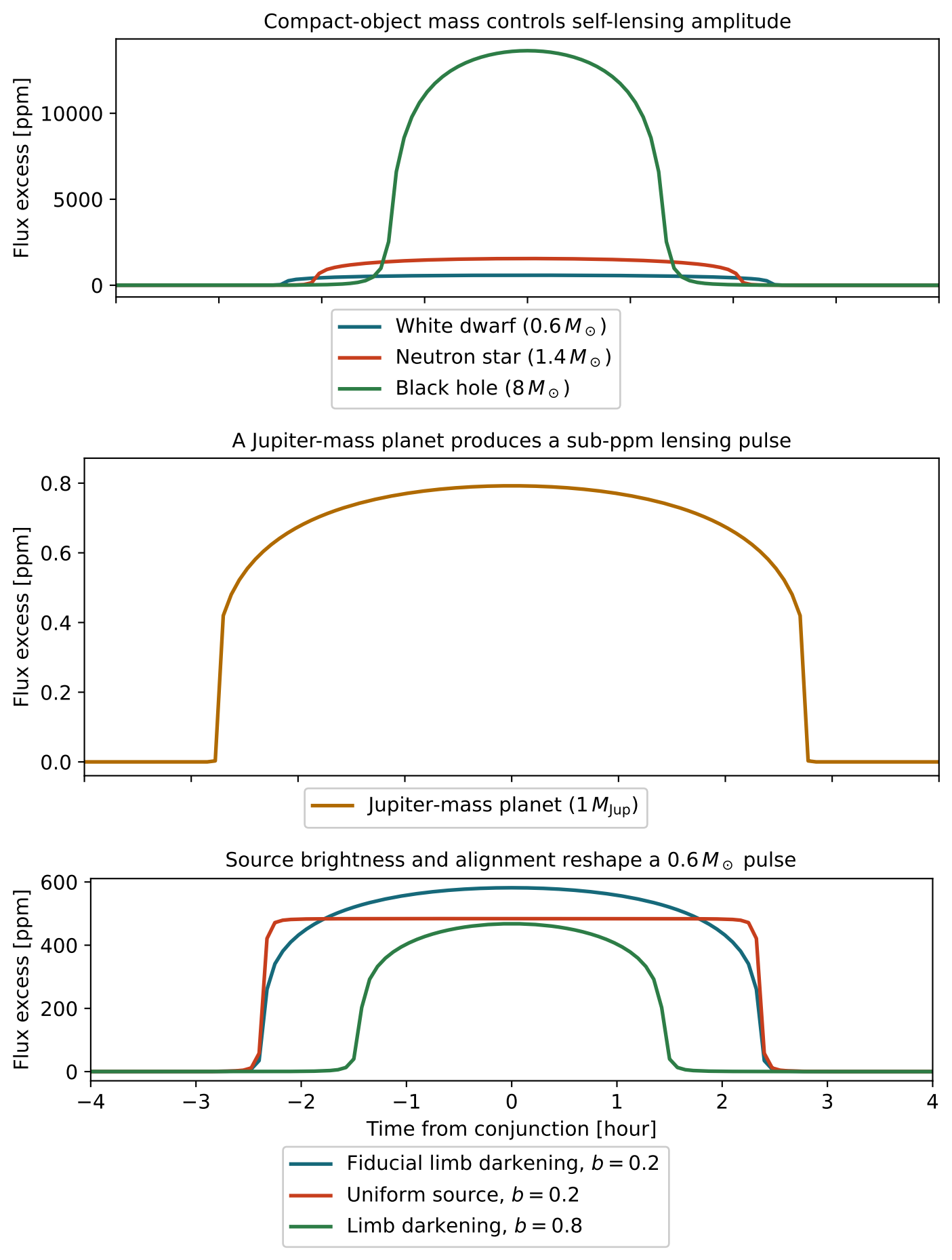
Compact-object mass controls self-lensing amplitude
Flux excess [ppm]
10000
5000
0
White dwarf (0.6 $M_\odot$)
Neutron star (1.4 $M_\odot$)
Black hole (8 $M_\odot$)
A Jupiter-mass planet produces a sub-ppm lensing pulse
Flux excess [ppm]
0.8
0.6
0.4
0.2
0.0
Jupiter-mass planet (1 $M_{\rm Jup}$)
Source brightness and alignment reshape a 0.6 $M_\odot$ pulse
Flux excess [ppm]
600
400
200
0
−4
−3
−2
−1
0
1
2
3
4
Time from conjunction [hour]
Fiducial limb darkening, $b = 0.2$
Uniform source, $b = 0.2$
Limb darkening, $b = 0.8$